

Rafael Ganascini de Moura

Backend Java Developer | Software Engineering Student

Belo Horizonte, MG, Brazil · Available for internship (6 hours/day)

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SUMMARY

Software Engineering student (6th semester, PUC Minas) with hands-on experience in Java/Spring Boot backend development on team projects with real clients. Experience building REST APIs, asynchronous messaging (RabbitMQ), caching with Redis, and load testing. Seeking a backend development internship to contribute to scalable solutions in a production environment.

PROJECTS

Biblioo (team project) · 2026

Brazilian literary community platform. Spring Boot backend built on hexagonal architecture with Redis, RabbitMQ, and OpenSearch, plus a Flutter mobile app. Cross-functional work in a team of 6, contributing to the backend.

- Developed the complete user module: JWT authentication (sign-up, login, refresh, and logout), social login with Google OAuth, Spring Security, a follow system, and integrations with OpenSearch and Cloudinary.
- Implemented the notification system integrating RabbitMQ and Firebase Cloud Messaging, with SSE and FCM adapters.
- Designed and implemented the platform's AI assistant on hexagonal architecture, using function calling to perform in-app actions from natural language, plus rate limiting, conversation history persisted in Redis, and anti-hallucination guardrails.
- Ran performance tests (load, spike, and stress) with k6, monitoring the application with Prometheus and Grafana and optimizing the backend code based on the results.

FieldFlow (individual project) · 2026 · github.com/RafaelMouraG/FieldFlow

Marketplace for agricultural services connecting rural producers to technical service providers.

- Independently designed and built a complete event-driven system: REST API in Python/FastAPI, two independent consumer workers, and a Flutter mobile app.
- Implemented fully asynchronous communication via RabbitMQ (topic exchange) with a Dead-Letter Queue and idempotency by event_id, ensuring resilience against failures and redelivery.
- Structured the backend in Clean Architecture by bounded context, with PostgreSQL, migrations via Alembic, JWT authentication, and automated tests with pytest.

Hortifruti Santa Luzia (team project, real client) · 2025

Business management system for a fresh produce retailer, including automated customer service. Cross-functional work in a team of 6, contributing to the backend.

- Developed a WhatsApp chatbot (WhatsApp Business) through which customers request and receive payment slips (boletos), invoices, and order updates, as well as the email notification system.
- Automated operational processes with integrated payment slip issuance and bank reconciliation, cutting time spent by over 90%: slip issuance from 2 hours/day to 10 minutes/day and reconciliation from 4 hours/week to 15 minutes/week.

EDUCATION

B.S. in Software Engineering, PUC Minas · Belo Horizonte, Brazil · Expected graduation: December 2027

6th semester in progress.

SKILLS

Languages: Java, Python, SQL, Dart (basic), C++

Backend: Spring Boot, Spring Security, Spring Data JPA, FastAPI, REST APIs, RabbitMQ, Redis, OpenSearch, JWT/OAuth

Databases: PostgreSQL, MySQL, JPA/Hibernate, SQLAlchemy, Alembic

Testing: JUnit, pytest, k6 (load testing)

Tools: Git, Docker, Docker Compose, Maven, Postman, Swagger/OpenAPI

Practices: Hexagonal architecture, Clean Architecture, event-driven architecture, load testing, Scrum

LANGUAGES

Portuguese: Native. · English: Advanced